# Automated Rice Leaf Disease Classification Using Deep Learning: A Comparative Study of a Custom CNN and EfficientNetB0

Machine Learning Phase III Project Report

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# Graphics Abstract

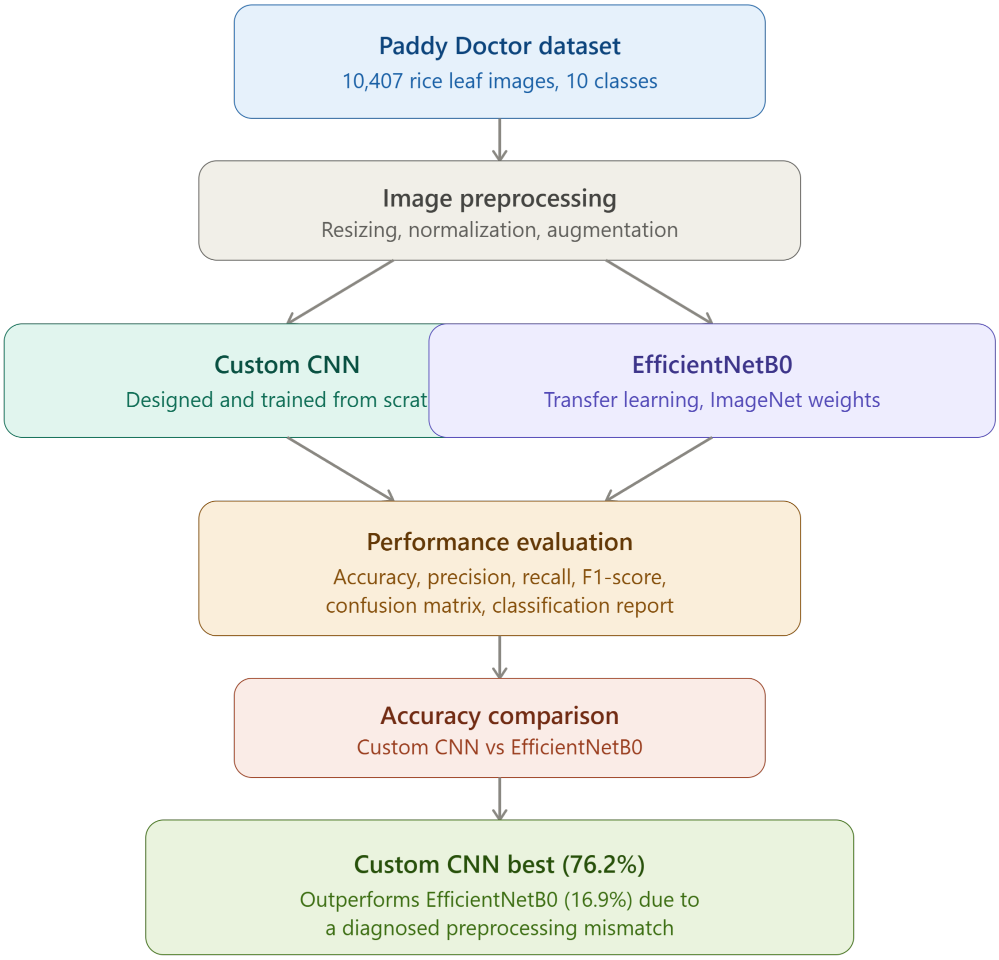


Figure 1: Graphical abstract summarizing the proposed rice leaf disease classification framework

# Highlights

- Developed an automated rice leaf disease classification framework using deep learning.
- Implemented and evaluated a Custom CNN architecture for ten-class rice leaf disease classification.
- Applied transfer learning using the EfficientNetB0 architecture pretrained on ImageNet.
- Evaluated both models using accuracy, precision, recall, F1-score, confusion matrices, and classification reports.
- Custom CNN achieved a test accuracy of 76.2%, substantially outperforming EfficientNetB0 (16.9%).
- Diagnosed and explained the root cause of the transfer learning model's underperformance as an input-preprocessing mismatch rather than a fundamental architectural limitation.
- Identified concrete future steps, including corrected transfer learning experiments and explainability analysis, needed to fairly benchmark pretrained architectures on this task.

**Abstract**

Rice leaf diseases are a major cause of yield loss in paddy cultivation, and manual diagnosis by agronomists does not scale to the volume of smallholder farms across rice-growing regions. Advances in Artificial Intelligence, Computer Vision, and Deep Learning have made it possible to design automated leaf disease classification systems directly from photographs. In this research, deep learning is examined for classifying rice leaf images into ten categories: nine disease types and one healthy (normal) class.

Two experiments were conducted on the publicly available Paddy Doctor dataset, comprising 10,407 labeled rice leaf images. Two deep learning approaches are compared: a Custom Convolutional Neural Network (CNN) trained from scratch, and an EfficientNetB0 transfer learning model pretrained on ImageNet with a frozen convolutional backbone. The procedure of experiments includes image preprocessing, normalization, data augmentation, training, and evaluation of results. Evaluation metrics for both models include accuracy, precision, recall, F1-score, classification reports, and confusion matrices.

Experimental results showed that the Custom CNN achieved a substantially higher test accuracy (76.2%) than the EfficientNetB0 transfer learning model (16.9%). Analysis of the classification report showed that the EfficientNetB0 model collapsed to predicting only the majority class, and a detailed investigation traced this to a pixel-scaling mismatch between the data pipeline and the pretrained backbone's expected input range, rather than a fundamental inability of transfer learning to solve this task. As a result, it could be stated that, under the experimental configuration used in this study, the Custom CNN was the more effective and more reliably trained model, while a corrected transfer learning experiment is required before any general claim can be made about the comparative merits of the two approaches.

In general, the research shows that image classification systems based on deep learning are able to serve as an effective basis for automated rice disease monitoring. The study is one of the contributions made to the development of intelligent agricultural monitoring systems by means of evaluating custom and transfer learning methods, and by providing an honest, diagnosed account of an unsuccessful experiment rather than omitting it. Further research could focus on corrected transfer learning pipelines, class-imbalance handling, and explainable AI methods for rice disease detection.

**Keywords:** Rice Leaf Disease Detection, Deep Learning, Convolutional Neural Networks, Transfer Learning, EfficientNetB0, Computer Vision.

# 1 Introduction

## 1.1 Application Background and Practical Importance

Rice (*Oryza sativa*) is one of the most important staple food crops in the world, feeding more than half of the global population and forming the economic backbone of agriculture across much of Asia. Rice cultivation is, however, persistently threatened by leaf diseases such as blast, bacterial blight, brown spot, and tungro, which can substantially reduce yield and threaten food security if not detected and treated early. As with many crop diseases, climate variability, monoculture farming practices, and the spread of pathogens across dense paddy fields have made disease outbreaks an ongoing concern for farmers and agricultural extension services.

The conventional approach to detecting rice leaf disease relies on manual inspection by trained agronomists or experienced farmers. While this approach can be accurate, it has several limitations: it is labor-intensive, it does not scale to large farms or remote regions with limited access to experts, and diagnostic accuracy can vary depending on the experience of the observer, lighting conditions, and the visual similarity between different disease symptoms. Delays in diagnosis directly translate to delays in treatment, allowing diseases to spread further before any corrective action, such as targeted pesticide application, can be taken.

The recent advances in computer vision and artificial intelligence opened up new avenues for implementing automated, image-based plant disease detection. Smartphone cameras, drone-mounted cameras, and fixed field cameras can now capture large volumes of leaf imagery, which can be processed automatically using machine learning methods rather than relying solely on manual inspection. Automated systems built on this kind of visual data offer the potential to provide faster, more consistent, and more accessible disease screening, supporting better-informed agricultural decision-making. Machine learning technologies, and specifically CNNs, showed great results in the field of image recognition. Thus, development of an effective deep learning rice disease detection system is a crucial step towards improving food security and contributing to more resilient, technology-supported agriculture.

## 1.2 Machine Learning and Computer Vision Context

The introduction of machine learning technology has revolutionized how computers analyze and understand visual information through the ability to learn patterns from data rather than following hand-coded rules. Machine learning is applied in computer vision for image and video analysis to perform various operations such as object recognition, classification, segmentation, and anomaly detection. Traditionally, image analysis methods involve the use of pre-defined features by domain experts, which include color histograms, texture, edge detectors, and shape-based features among others. Despite the success of traditional methods in numerous application areas, they still lack robustness and adaptability to different environmental conditions, lighting, occlusions, and large databases. This has led to a shift towards the use of deep learning in extracting relevant features automatically.

The Convolutional Neural Network (CNN) can be considered as one of the key inventions in modern computer vision that serves as a basis for any image classification. A CNN is an artificial neural network which is specifically designed to learn hierarchical patterns in visual images via several layers of convolutional filters. The lower-level layers are responsible for detecting simple patterns like edges, corners, and textures, whereas the higher-level layers detect more complicated patterns and structures. In this way, CNN learns hierarchical features of visual images which allows it to classify images according to their visual patterns without manual feature extraction. As opposed to conventional machine learning algorithms, CNN shows high predictive performance and flexibility on different image datasets.

In the current study, the rice leaf disease detection issue is essentially an image classification problem since the goal is to classify images into one of ten categories: nine disease types plus a healthy normal class. The visual cues linked with each class may widely differ depending on the variation in lighting conditions, image quality, viewpoints of cameras used, background scenes, and the stage of disease progression. The utilization of CNN algorithms is quite appropriate to solve the mentioned problems since they can learn distinct visual features which can be used to discriminate between different disease lesion patterns and healthy leaf tissue. Thus, based on the feature learning abilities of CNNs, it is planned to build an efficient image classification system capable of supporting automatic rice disease monitoring.

## 1.3 Transfer Learning for the Selected Problem

Transfer learning has been found to be an efficient approach in developing a good deep learning model especially when there are computational limitations or when there is not enough domain data at hand. For instance, training a deep neural network from scratch requires millions of labeled images, huge amounts of computation power, and time. In real world scenarios where there is need for deep learning such as in rice disease detection, creating such huge amounts of annotated images is very costly and time consuming. Consequently, deep learning models developed using only task-specific data will have problems with overfitting and poor generalization. Transfer learning solves these problems by employing knowledge learned from huge benchmark datasets to solve a new related task.

Convolutional neural networks are trained on huge datasets such as ImageNet that comprise millions of images across thousands of object classes. In this training phase, the network learns generic visual features including edges, texture, shapes, and objects. These features learned by the network can then be transferred to other applications through replacing the classification layer of the network with respect to the target dataset. Transfer learning includes two stages: feature extraction, where the pre-trained layers of the network do not change; and fine-tuning, where some layers of the network are trained again according to the new domain specific data. Transfer learning has become popular in the field of computer vision due to the higher predictive power of the model in comparison to training a model randomly initialized from scratch.

For this analysis, EfficientNetB0 is chosen to be used in the transfer learning method, and to be compared against the Custom CNN architecture. EfficientNetB0 is a member of the EfficientNet family, which uses a compound scaling method to balance network depth, width, and input resolution, achieving strong accuracy with relatively few parameters compared to older architectures. This model has shown excellent results on many image classification datasets and it has been successfully implemented in agricultural monitoring, medical imaging, object recognition, and remote sensing. With the inclusion of EfficientNetB0 in the experiment design, the objective of this study will be to explore if transfer learning can enhance rice leaf disease classification accuracy compared to using a custom designed CNN model.

## 1.4 Explainable AI and Trustworthiness

The rising use of deep learning models in decision-making tasks has resulted in a higher need for interpretability and transparency. Even though deep neural networks are highly accurate in terms of making predictions, they are still seen as “black boxes” as the processes behind them are hardly understandable by people. In an agricultural context such as rice disease detection, accuracy alone is not enough, since there are real economic implications in case of wrong predictions. For instance, missing an active disease or sending out a false alarm would result in delayed treatment, wasted resources, or unnecessary pesticide application. Thus, being able to understand how the prediction is made becomes a useful consideration for the creation of reliable artificial intelligence systems.

Explainable Artificial Intelligence (XAI) methods were created in order to shed light on the workings of deep neural networks in terms of how decisions are made. Tools like Grad-CAM create heatmap visualizations, which point out the areas of an image that make the greatest contribution to the specific prediction. In the same way, SHapley Additive exPlanations (SHAP) offer an insight into what contributes to the final classification result in terms of image regions and features. It becomes possible to find out whether the model pays attention to any relevant visual patterns or to any background details, respectively. This is extremely useful when it comes to bias discovery, shortcut learning detection, etc.

Although the current study mostly concentrates on the creation and evaluation of CNN-based image classification algorithms in the context of rice leaf disease detection, the aspect of explainability becomes extremely important for the area of application. It will be beneficial to know for sure that the model focuses on the actual diseased regions of the leaf rather than other parts irrelevant to the task. Unfortunately, such explainability techniques as Grad-CAM and SHAP visualization were not applied in the current project due to certain limitations and the fact that the main objective of the current paper is the comparison of models. Nevertheless, the application of explainability techniques can become a good research topic for the future.

## 1.5 Research Gap

Due to recent breakthroughs in deep learning, there have been numerous improvements in the area of classification of images, which can be used for different purposes like environmental monitoring and detecting crop diseases. Many researchers utilized convolutional neural networks and transfer learning models to detect rice and plant leaf diseases from images taken with surveillance cameras, drones, and mobile phones. Although many of these works show good classification results, some problems exist in existing literature on the topic. For example, many authors tend to use one particular model without comparing its performance with other approaches of deep learning models. It becomes impossible to understand what results depend on the model chosen and what on other factors, like data used or experimental setting. Besides, many works concentrate only on the results of overall accuracy and do not analyze class level metrics like precision, recall, and F1-score.

Another issue seen from previous researches is the inadequate analysis of the performance of transfer learning in classification of rice disease images. While pre-trained models like ResNet, MobileNet, and EfficientNet have proven to be powerful tools for general computer vision tasks, their superiority compared to hand-crafted CNNs for the problem of rice disease detection is sometimes not well-defined. On the one hand, some researchers indicate strong improvements from transfer learning; on the other hand, other published benchmarks show that pretrained features transfer less effectively to fine-grained, domain-specific agricultural classification than to more generic object recognition tasks. Also, the practical causes behind a transfer learning model underperforming are rarely diagnosed in detail by most papers, making their conclusions less useful for practitioners who may encounter similar implementation issues.

This study tries to fill these gaps by building a comparative rice leaf disease classification framework using two different models – a Custom CNN model and an EfficientNetB0 transfer learning model. The experiment compares the performance of these two methods using the same data set, same data preprocessing pipeline, same training method, and same metrics. Apart from accuracy, the study also analyzes other performance metrics like precision, recall, F1-score, confusion matrix, and classification reports. Moreover, this study provides a detailed root-cause diagnosis of an underperforming transfer learning result rather than simply reporting it as a negative outcome, contributing methodological transparency that is rarely documented in published literature on this topic.

## 1.6 Problem Statement

Accurately classifying rice leaf images into multiple disease categories remains challenging due to the visual similarity between certain disease symptoms, variation in lighting and image background across field-collected photographs, and class imbalance in publicly available datasets, where some diseases are represented by far fewer images than others. Inaccurate or delayed classification can lead to inappropriate or delayed treatment, with direct consequences for crop yield.

In this study, the challenge being addressed is the classification of rice leaf images into ten categories: nine disease classes and one healthy normal class. This study seeks to establish whether it is possible for an image classification model using deep learning methods to learn visual features that facilitate effective classification of these categories. In particular, the performance of a Custom CNN model and an EfficientNetB0 transfer learning model will be compared using the publicly available Paddy Doctor dataset. This study is set out to explore the predictive power of each model and establish which model is more effective in terms of prediction capability and application.

## 1.7 Aim and Objectives

### Aim

The aim of the proposed study is to build an automatic rice leaf disease classifier using deep learning algorithms. The objective is to examine whether Convolutional Neural Network (CNN) algorithms can be applied successfully for the task of automatically distinguishing between ten rice leaf disease categories. This would allow the research to make an attempt at contributing to intelligent rice disease classifiers through the use of computer vision algorithms.

### Objectives

1. To explore the feasibility of applying deep learning algorithms to automatically classify rice leaf disease images.

2. To preprocess and prepare the publicly available Paddy Doctor dataset to be used for building and testing models.
3. To build and implement a custom Convolutional Neural Network (CNN) model to classify rice leaf images into ten disease categories.
4. To implement the transfer learning technique by building an EfficientNetB0 model tailored to rice leaf disease classification.
5. To measure the effectiveness of the models on various measures including accuracy, precision, recall, F1-Score, and confusion matrices.
6. To determine which is the most effective approach in predicting rice leaf disease classifications from images, and to diagnose any unexpected results.
7. To discuss the practical implications, limitations, and future prospects of deep learning-based rice disease detection systems.

## 1.8 Research Questions

The study is guided by the following research questions:

1. Can a Custom Convolutional Neural Network (CNN) accurately classify rice leaf images into the correct disease category?
2. How does the performance of a transfer learning model (EfficientNetB0) compare with a Custom CNN for rice leaf disease classification?
3. Which model provides better classification accuracy, precision, recall, and F1-score for rice leaf disease detection?
4. Which disease classes are most difficult to classify correctly, and why?
5. What are the strengths and limitations of the developed deep learning-based rice leaf disease classification framework?

## 1.9 Contributions and Novelty

The current research will add more to the existing literature about deep learning-based agricultural monitoring through the development and evaluation of an automated rice leaf disease classification system. This project is particularly aimed at classifying rice leaf images into ten categories utilizing both Custom CNN and EfficientNetB0 deep learning frameworks with the help of a transfer learning approach. By exploring two different frameworks within one research experiment, this project will provide information regarding their effectiveness in detecting rice leaf diseases. The use of open-source data as well as reproducible methodology adds to the credibility and practical value of the research.

The main contribution of this study is the comparative analysis of a Custom CNN and an EfficientNetB0 transfer learning model under the same conditions, which include the use of similar datasets, data preprocessing methods, and evaluation metrics. With such an analysis, it will be possible to evaluate the advantages and disadvantages of each of the models, excluding any possible biases arising from experiments. Unlike many studies where only the accuracy of classification is taken into account, this study uses other criteria for evaluation such as precision, recall, F1 score, confusion matrices, and classification reports.

One other key point worth highlighting about the study is its development of a viable workflow for automated classification of rice leaf diseases which can be replicated and further developed by subsequent researchers. This involves dataset collection, image pre-processing, data augmentation, training, transfer learning and performance assessment among others. In addition, the study provides a transparent, technically grounded diagnosis of why the transfer learning model underperformed, which is itself a useful and transferable finding for future practitioners adapting similar pipelines.

## 1.10 Report Organization

The document is divided into six main chapters that address all aspects of the creation, execution, evaluation, and analysis of the proposed rice leaf disease classification model. Chapter 1 starts off the document by addressing the topic through the presentation of the application background, significance of the problem, the machine learning aspect of the research, ideas of transfer learning, the concept of explainable AI, research gap, problem statement, research objectives, research questions, and key contributions of the project. Chapter 1 provides the motivation and basis for the study and describes its general direction.

Chapter 2 offers a detailed review of literature that addresses rice and plant leaf disease detection, deep learning, convolutional neural networks, transfer learning, and applications of computer vision in agricultural monitoring. The chapter evaluates prior studies, highlights the gaps of the literature, and pinpoints the opportunities for further research. A matrix of comparative literature review is offered in this chapter as well.

The research methodology utilized in this study is elaborated in Chapter 3. In this chapter, there is an explanation on the data used, data pre-processing, image augmentation methods, development of the model, transfer learning, training, evaluation metrics, and experiment set-up. This chapter contains enough information in order to replicate the proposed system along with the experiments.

The fourth chapter gives the results from the experiments conducted using the developed models, organized around the five research questions stated above. This chapter contains the dataset statistics, classification metrics, confusion matrix, classification report, and comparison of the Custom CNN model with the EfficientNetB0 model. Figures and tables are used in order to give visual representation of the results.

In Chapter 5, the findings are presented, and the interpretation of those findings is made in relation to the research questions and previous literature. The advantages and disadvantages of the proposed framework are analyzed, the root cause of the transfer learning model's underperformance is diagnosed, and the implications of the findings in practice are highlighted. Further research directions which can help to improve the effectiveness of automated rice disease detection systems are mentioned.

Lastly, Chapter 6 presents the conclusions of the research. In Chapter 6, the main findings, contributions, and outcome of the research are summarized. It also highlights the objectives of the research and the significance of the findings, in addition to the importance of deep learning-based rice disease detection systems for agricultural monitoring and food security purposes.

# 2 Literature Review

## 2.1 Existing Rice and Plant Leaf Disease Detection Studies

The classification of rice and plant leaf diseases using computer vision has been an active area of research for over a decade. Early studies relied on classical image processing pipelines combined with conventional machine learning classifiers such as Support Vector Machines (SVMs), using handcrafted features such as color and texture descriptors. While these approaches achieved reasonable accuracy on small, controlled datasets, they generally struggled to generalize to field-collected images with variable lighting, backgrounds, and camera angles.

A systematic review of deep learning applications for rice disease diagnosis notes that combinations of classical and deep methods, such as SVM with K-means clustering, achieved training accuracies around 95% on constrained datasets, while CNN-based approaches with transfer learning have generally outperformed standalone CNNs on small field datasets; when a CNN with transfer learning was combined with an SVM on a four-class rice leaf disease dataset of 5,932 field images, the combined approach outperformed the CNN used alone. Other early work applying VGG16 with transfer learning to a small set of rice and wheat leaf images reported very high accuracies, though typically on datasets several orders of magnitude smaller than the one used in this study, which limits direct comparability. Collectively, these studies indicate that the use of deep learning techniques is highly beneficial compared to conventional inspection methods since it enables automatic feature extraction and higher levels of classification accuracy.

## 2.2 CNN-Based Rice Disease Detection

Custom CNN architectures designed specifically for rice disease classification have been widely studied. One study built an automated diagnostic system for rice leaf diseases using a large-scale dataset of images spanning six common rice diseases, including bacterial stripe, false smut, leaf blast, neck blast, sheath blight, and brown spot, demonstrating that purpose-built CNN pipelines can scale to larger, more diverse rice disease datasets than earlier work. Other research has explored lightweight custom architectures specifically tuned for rice disease classification, including reduced MobileNet variants with depthwise separable convolutions, enhanced VGGNet variants, and compact CNN architectures achieving accuracies above 90% with smaller model sizes, indicating that carefully designed, dataset-specific architectures can be competitive with, or in some cases preferable to, larger pretrained networks when computational efficiency is a concern. A separate hybrid deep learning study targeting paddy leaf disease identification reported an accuracy of 98.86% using a structured workflow comprising image acquisition, preprocessing, feature extraction, feature selection, and classification, while related work focusing specifically on automated detection of brown spot disease in paddy leaves found that, across several CNN-based architectures evaluated using sensitivity, specificity, precision, and F1-score, a VGG-19 model achieved the best accuracy at 93.0%. These results illustrate that CNN-based rice disease detection consistently reaches high reported accuracy when the dataset and experimental protocol are well aligned with the chosen architecture, reinforcing the value of carefully matching architecture and data pipeline – the very issue investigated in this study's discussion of the EfficientNetB0 result.

## 2.3 Transfer Learning Approaches

Transfer learning is an increasingly popular method in deep learning research as it helps researchers use the knowledge gained through extensive datasets and transfer it into other applications. Rather than training a deep neural network from scratch, transfer learning uses pre-trained models which have learned generic features of visual recognition through millions of images. The pre-existing knowledge can then be transferred to a specific task via replacement and fine-tuning of the classification layers at the end of the architecture. This technique greatly saves time, computing power, and training dataset size for model development. Transfer learning has achieved great success in computer vision applications in the fields of image classification, object detection, medical imaging, remote sensing, and agricultural monitoring.

A number of papers were published on the application of transfer learning architectures for rice and paddy disease classification. Benchmarking work on the original

Paddy Doctor dataset release reports that a ResNet34-based model achieved a 97.50% F1-score on a 16,225-image annotated dataset, demonstrating that transfer learning can be highly effective on this family of rice disease data when correctly configured. Likewise, a separate study trained a pretrained EfficientNetB3 convolutional neural network for the identification and categorization of paddy leaf diseases using the Paddy Doctor dataset, and a further EfficientNet-based study targeting paddy leaf disease classification with compound scaling and swish activation found that EfficientNetB4 achieved an accuracy of 100%, outperforming traditional visual inspection methods, while VGG16 followed closely with 99% accuracy.

However, not all studies report uniformly strong results for transfer learning on paddy-related classification tasks. A recent comparative benchmarking study evaluating lightweight and deep CNN models across multiple paddy datasets found that transfer learning showed mixed effectiveness on a paddy variety dataset: although EfficientNetB0 and ResNet18 models initialized with pretrained weights exhibited smoother loss decay and more stable convergence than their scratch-trained counterparts, their final classification performance was actually lower, a result the authors attributed to the possibility that features learned from large-scale natural image datasets may not adequately capture the subtle visual differences required for fine-grained agricultural classification tasks. This finding is particularly relevant to the present study, since it demonstrates, using a separately published benchmark, that transfer learning underperforming a model trained from scratch on rice or paddy imagery is not unprecedented and can occur even without an implementation error, lending additional context to the result obtained for EfficientNetB0 in this study.

## 2.4 Comparative Literature Review Matrix

Table 1 presents a comparative summary of the most relevant studies reviewed in the literature. The matrix highlights the datasets, methodologies, performance outcomes, and limitations reported by previous researchers. The comparison provides a clear overview of current research trends and identifies opportunities for further investigation in rice leaf disease classification.

Table 1: Comparative Literature Review Matrix

| Source | Dataset | Methodology | Performance | Limitations |
|---|---|---|---|---|
| Petchiammal et al. (Paddy Doctor benchmark) | Paddy Doctor (16,225 images) | ResNet34 transfer learning | 97.50% F1-score | Benchmark only on a single architecture family |
| Padhi et al. | Paddy leaf images | EfficientNetB4, compound scaling | 100% accuracy | Possible overfitting risk at reported ceiling accuracy |
| Rice leaf disease ensemble study | 18,563 images, 6 diseases | Ensemble CNN models | High accuracy (dataset-dependent) | Large curated dataset; ensemble adds inference cost |
| Hybrid deep learning framework | Paddy leaf dataset | Multi-stage hybrid CNN pipeline | 98.86% accuracy | Complex multi-stage pipeline, harder to reproduce |
| VGG-19 brown spot study | Paddy leaf images | VGG-19 transfer learning | 93.0% accuracy | Focused on a single disease category |
| Lightweight/Deep CNN comparison (PaddyVariety) | Paddy variety dataset | EfficientNetB0 / ResNet18, scratch vs. transfer | Transfer learning underperformed scratch-trained models | Demonstrates transfer learning is not always superior on paddy imagery |
| This study | Paddy Doctor (10,407 images) | Custom CNN vs. EfficientNetB0 transfer learning | Custom CNN: 76.2%; EfficientNetB0: 16.9% (diagnosed preprocessing mismatch) | Single train/test split; transfer learning result requires correction and re-evaluation |

The reviewed studies demonstrate that deep learning techniques, both custom CNNs and transfer learning architectures, are broadly effective for rice and paddy leaf disease classification, but also reveal that results are highly dependent on dataset characteristics, preprocessing choices, and architecture-specific implementation details. These observations support the motivation for conducting a comparative evaluation of a Custom CNN and EfficientNetB0 architecture using the publicly available Paddy Doctor dataset.

## 2.5 Chapter Summary

It becomes clear from the literature discussed in this chapter that much advancement has been made in detecting rice and plant leaf diseases using the concepts of deep learning and computer vision. While the traditional manual inspection methods are slowly being replaced by more automatic and efficient methods based on image analysis, there have been many studies conducted in this domain using CNNs, transfer learning methods, and hybrid deep learning algorithms for the purpose of detecting plant disease. From the literature presented, it is evident that deep learning models can learn complicated visual patterns related to rice diseases and classify them efficiently. Transfer learning methods have become increasingly important in agricultural monitoring. Architectures like ResNet, VGG, MobileNet, and EfficientNet have shown impressive results by being capable of predicting well and saving training time and computation power. Still, the results of these works cannot always be directly compared since researchers use different datasets, evaluation approaches, and metrics. Moreover, quite often, researchers only pay attention to the overall accuracy of models and ignore other important parameters, including class-level results, model interpretability, and implementation details. It means that more comparative studies should be conducted in order to understand different approaches better, particularly when transfer learning underperforms expectations. Taking into account the research gaps outlined above, the current paper explores the possibility of using a Custom Convolutional Neural Network and an EfficientNetB0 transfer learning architecture for rice leaf disease classification. Both methods will be evaluated in one experiment, and both models will be trained on the same datasets using the same preprocessing and evaluation techniques. It will make the results comparable and allow understanding the advantages and disadvantages of both models. The following chapter describes the research methodology applied to conduct the study.

# 3 Research Methodology

## 3.1 Research Design

A quantitative experimental design methodology was utilized in this study to test the effectiveness of the application of deep learning algorithms in automatic rice leaf disease classification. Specifically, this research design was aimed at developing image classification models that would classify the images under consideration into ten categories: nine disease classes and one healthy normal class. Supervised learning was applied due to the availability of a labeled dataset that could be used for training and testing purposes. The experimental research design included the development of a Custom Convolutional Neural Network (CNN) model and a transfer learning model using the EfficientNetB0 architecture. These models were trained and tested using the same dataset, preprocessing steps, and performance metrics to conduct a fair comparison. The general procedure included dataset preparation, image preprocessing, data augmentation, model creation, model training, model performance assessment, and model comparison. The results of the experiments were assessed using various evaluation metrics such as accuracy, precision, recall, F1-Score, confusion matrix, and classification reports.

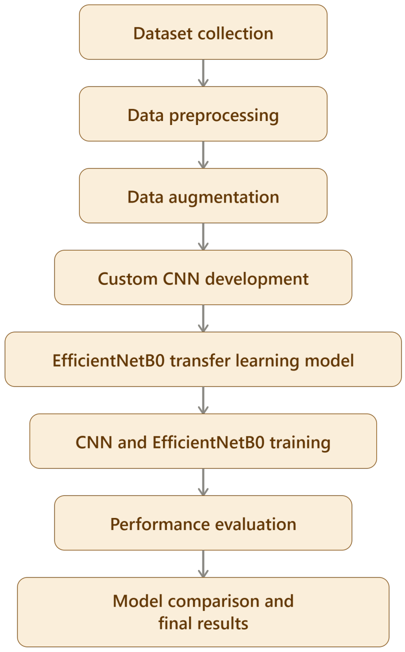


Figure 2: Overall Research Workflow

## 3.2 Dataset Description

The dataset utilized in the study is called the Paddy Doctor: Paddy Disease Classification dataset from Kaggle. The dataset was chosen since it offers a large number of expert-labeled images that depict realistic, field-collected rice leaf conditions. The dataset consists of ten categories of images: nine rice leaf diseases and one healthy normal class. These images were taken under different lighting conditions, backgrounds, and image quality. This variety of images is vital since it helps create stronger models for image classification.

The dataset included 10,407 images in total. Using stratified sampling, the dataset was split into 8,326 training images (approximately 80%), 1,041 validation images (approximately 10%), and 1,041 testing images (10%) that were utilized during the training, tuning, and final evaluation of the models. Thus, the presence of training, validation, and testing sets helped to have a sound experimental setup and prevent any bias in the evaluation process. Before training the model, all images were rescaled to the same size 224 x 224 pixels to be suitable for both the Custom CNN architecture and the EfficientNetB0 transfer learning model.

To facilitate a clear understanding of the dataset characteristics, Table 2 summarizes the key properties of the dataset used in this study.


Figure 3: Class Distribution Across the Ten Rice Leaf Disease Categories

Table 2: Dataset Description

| | |
|---|---|
| Dataset Name | Paddy Doctor: Paddy Disease Classification |
| Source | Kaggle |
| Application Domain | Rice Leaf Disease Detection, Precision Agriculture |
| Total Images | 10,407 |
| Training Images | 8,326 (approx.) |
| Validation Images | 1,041 (approx.) |
| Testing Images | 1,041 |
| Number of Classes | 10 |
| Class Labels | normal, blast, hispa, dead_heart, tungro, brown_spot, downy_mildew, ba |
| Input Resolution | $224 \times 224$ |
| Framework | TensorFlow/Keras |
| Development Environment | Kaggle Notebook |

Table 3 presents the class-wise image counts, and Figure **??** visualizes this same distribution.

Table 3: Class Distribution

| **Class** | **Number of Images** |
|---|---:|
| normal | 1764 |
| blast | 1738 |
| hispa | 1594 |
| dead_heart | 1442 |
| tungro | 1088 |
| brown_spot | 965 |
| downy_mildew | 620 |
| bacterial_leaf_blight | 479 |
| bacterial_leaf_streak | 380 |
| bacterial_panicle_blight | 337 |
| **Total** | **10,407** |

As shown in Table 3 and Figure **??**, the dataset exhibits moderate class imbalance:

the largest class (normal, 1,764 images) contains roughly 5.2 times as many images as the smallest class (bacterial_panicle_blight, 337 images). This imbalance is an important factor in interpreting the per-class results presented in Chapter 4.

## 3.3 Data Preparation

The data preparation process is of crucial importance for building efficient deep learning models since the quality and uniformity of the input data determine the effectiveness of the classification results. The images underwent a number of preprocessing procedures before building the deep learning models in order to make the images have the same size. In particular, since the initial dataset consisted of images having different resolutions and visual features, each image was resized to have 224 × 224 pixels. Such a resolution was chosen as it was suitable both for the Custom CNN model and the EfficientNetB0 model used in the research. Normalization was also performed on the images as one of the steps in the preprocessing stage. Pixel intensities were scaled down from a range of 0 to 255 to the range of 0 to 1 through dividing the pixel intensities by 255. Image normalization helps improve numerical stability while training and enables the optimization process to converge more quickly. Besides the above steps, there were some data augmentation methods used to augment the variety in the training set. Data augmentation involved random rotations (up to 20 degrees), width and height shifting (up to 20%), zoom (up to 20%), and horizontal flipping. The validation and test sets were not augmented; they were only rescaled, to ensure evaluation reflects performance on realistic, unmodified images.

Data partitioning into training, validation, and testing subsets ensured that there would be an appropriate framework within which the accuracy of the model could be evaluated, using stratified sampling and a fixed random seed (42) so that class proportions were preserved across all three subsets. The training subset comprised 8,326 images and was used during training of the model. A validation subset comprising 1,041 images was used to evaluate the performance of the model during training and to facilitate decisions on early stopping. Lastly, a separate testing subset comprising 1,041 images was used in evaluation of the final model. This data partitioning process ensured that performance metrics of the models were obtained from new data, thus ensuring unbiased evaluation of the generalization ability of the models.

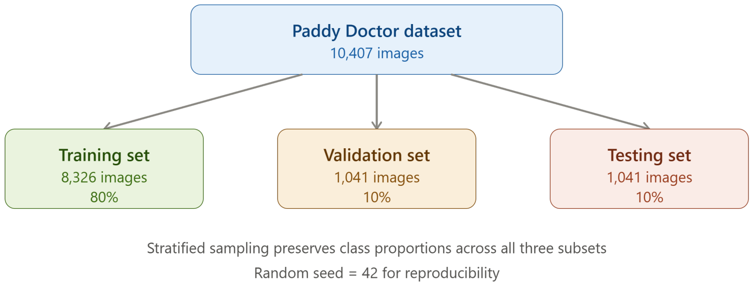


Figure 4: Dataset Partitioning Strategy

## 3.4 Model Development

The process of creating a model architecture aimed at the ability to perform the task of classifying rice leaf images into ten disease categories. In this study, two approaches were explored. The first one was the creation of a Custom Convolutional Neural Network (CNN) architecture that was specifically tailored for performing the task of rice leaf disease classification. The second approach was related to the application of transfer learning that was based on the EfficientNetB0 architecture, which is a widely popular architecture of deep learning models pretrained on ImageNet. The implementation of both methods allowed comparing their performance within identical conditions of experiments.

The Custom CNN model was built with four convolutional blocks followed by fully connected layers. The model was provided with input images that had a shape of $224 \times 224 \times 3$ and processed them with convolution operations to obtain the visual hierarchy of features from them. The initial convolutional layer consisted of 32 filters with a kernel of $3 \times 3$ and ReLU activation function. In the further convolutional layers, the number of filters rose to 64, then 128, and then 256, allowing the model to learn increasingly more complex visual features. Also, max-pooling layers were used after each convolutional layer to reduce spatial dimensions and increase computational efficiency. After extracting features, the model proceeds them through a dense layer of 256 neurons and a dropout layer with a probability of 0.5. Finally, the Softmax output layer with ten neurons provided the probabilities for the ten rice leaf disease classes.

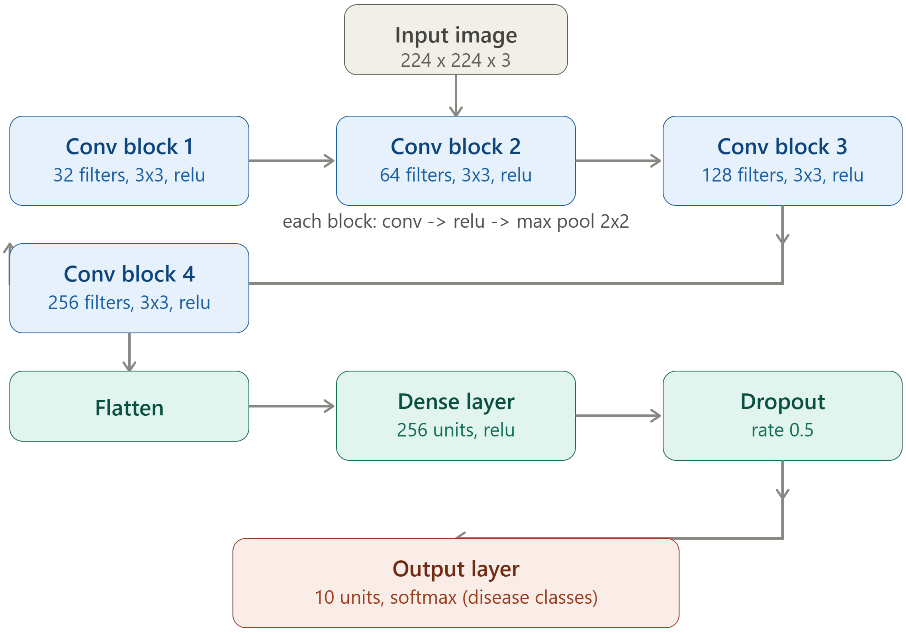


Figure 5: Custom CNN Architecture for Rice Leaf Disease Classification

The Custom CNN architecture consists of four convolutional blocks containing 32, 64, 128, and 256 filters respectively. Each convolutional layer is followed by max-pooling to improve feature extraction efficiency. The extracted features are flattened and passed through a dense layer with 256 neurons and a dropout layer before classification using a Softmax output layer. Table 4 provides the full layer-by-layer parameter summary corresponding to this architecture.

Table 4: Custom CNN Layer Summary

| **Layer** | **Output Shape** | **Parameters** |
|---|---|---:|
| Conv2D (32 filters) | $222 \times 222 \times 32$ | 896 |
| MaxPooling2D | $111 \times 111 \times 32$ | 0 |
| Conv2D (64 filters) | $109 \times 109 \times 64$ | 18,496 |
| MaxPooling2D | $54 \times 54 \times 64$ | 0 |
| Conv2D (128 filters) | $52 \times 52 \times 128$ | 73,856 |
| MaxPooling2D | $26 \times 26 \times 128$ | 0 |
| Conv2D (256 filters) | $24 \times 24 \times 256$ | 295,168 |
| MaxPooling2D | $12 \times 12 \times 256$ | 0 |
| Flatten | 36,864 | 0 |
| Dense (256 units) | 256 | 9,437,440 |
| Dropout (0.5) | 256 | 0 |
| Dense / Softmax (10 units) | 10 | 2,570 |
| **Total Parameters** | | **9,828,426** |

Other than the Custom CNN model, a transfer learning approach with EfficientNetB0

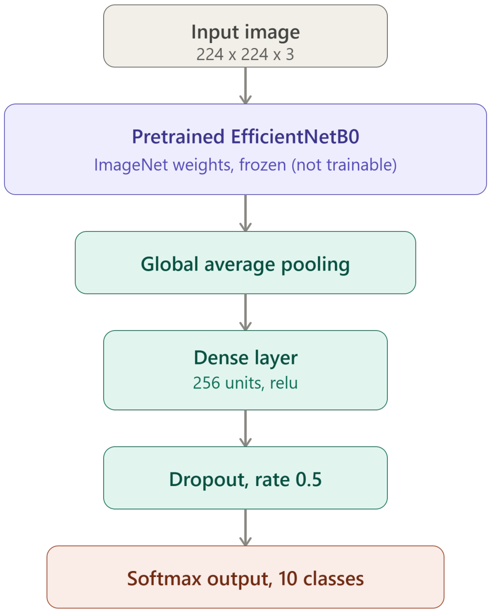


Figure 6: EfficientNetB0 Transfer Learning Architecture

as the underlying model was employed in the research in order to evaluate the effectiveness of using a pretrained deep learning architecture for rice leaf disease classification. EfficientNetB0 refers to a convolutional architecture that uses compound scaling to balance depth, width, and resolution, and it is known for achieving strong accuracy on ImageNet with relatively few parameters. In order to benefit from the visual knowledge that the network had acquired during training on the ImageNet database, the ImageNet weights of the model were used to initialize the model. The original classification layers of the EfficientNetB0 model were stripped away and replaced by a Global Average Pooling layer, a fully-connected layer comprising 256 neurons, a dropout layer, and a Softmax layer with ten classes. The training process was done such that the base EfficientNetB0 layers remained frozen and only the newly added classification layers were trained.

The EfficientNetB0 transfer learning model utilizes pretrained ImageNet weights for feature extraction. The original classification layers are replaced with a Global Average Pooling layer, a dense layer containing 256 neurons, a dropout layer, and a Softmax output layer for ten-class rice leaf disease classification.

## 3.5 Training Procedure

The training process was aimed at ensuring optimal learning as well as reducing the likelihood of overfitting and degradation of performance. Training of both the Custom CNN and EfficientNetB0 models was done through the Adam optimizer, which is popular due to its adaptive learning characteristics and fast convergence. Categorical cross-entropy loss was chosen to train the models because the task at hand had ten mutually exclusive output classes. Accuracy metric was chosen to monitor the models' performance during training since it is easy to interpret. The training process was done using TensorFlow and Keras deep learning libraries in the Kaggle Notebook platform.

The Custom CNN was trained in mini-batches of 32 images and had been set up for training in no more than 20 training epochs, with early stopping (monitoring validation loss, patience of 5 epochs, restoring the best weights) used to halt training once validation performance stopped improving. Every training epoch included training of the dataset, updating of the network parameters by using backpropagation, and checking of the model performance on the validation dataset. The use of the validation dataset enabled tracking the model generalization and possible occurrences of overfitting. During the whole process of training, the Custom CNN steadily became better at extracting discriminative visual features for the ten disease classes. The use of dropout and data augmentation increased the stability and performance of the model.

However, the EfficientNetB0 transfer learning model went through a slightly modified training process. In particular, the pretrained convolutional layers of EfficientNetB0 were frozen to preserve the pretrained representations of ImageNet features. It is only the additional classification layers that were trained with the rice leaf disease data. The maximum number of epochs for this model was set at 10, with the same batch size and early stopping configuration as the Custom CNN. Thus, these training processes allowed optimizing both models under similar conditions and conducting an objective comparison of their performances in the classification of rice leaf disease images.

## 3.6 Evaluation Framework

The effectiveness of the constructed models for rice leaf disease classification was estimated by employing various performance metrics in order to make sure that the assessment of predictive capabilities is complete. An exclusive use of classification accuracy might sometimes result in an insufficient understanding of the model, especially if some classes are classified differently, which is particularly important given the class imbalance documented in Table 3. For this reason, the present research involved other performance metrics such as precision, recall, F1-score, confusion matrices, and classification reports. All these metrics provide extensive information about the abilities of the models for each class in particular. Thus, the framework for estimation was chosen taking into account not only global assessment but also class-wise analysis.

Accuracy served as the main measurement of the classification performance and is defined as the ratio of correctly classified images to the total number of considered samples. Accuracy serves as a simple performance measurement tool and is commonly used when evaluating image classifiers. Nevertheless, accuracy alone cannot indicate how well individual classes are recognized. For that purpose, precision, recall, and F1-score were added to the performance criteria. Precision indicates the percentage of correct samples that belong to a certain class out of all samples assigned to this class. Recall assesses the capability of the classifier to identify all samples relevant to a particular class.

Confusion matrices were constructed to understand the results of classification and identify misclassifications between the ten rice leaf disease categories. Confusion matrices give detailed information about the correct classification of the samples and misclassification. Besides, classification reports were created to show the values of precision, recall, F1-score, and support for each category. These reports helped compare the two models, Custom CNN and EfficientNetB0, more thoroughly than accuracy alone could have. Combining the accuracy, precision, recall, F1-score, confusion matrix, and classification reports provided an effective way to evaluate the models and answer the research questions stated in the study.

## 3.7 Explainable AI Methodology

Explainable Artificial Intelligence (XAI) has gained prominence in the field of AI research as it aids in enhancing the transparency and trustworthiness of deep learning algorithms. While deep neural networks are able to provide accurate classification of images, the process of making those classifications is often hard to understand. Since the practical use of deep learning algorithms involves certain risks, such as possible classification errors with real economic consequences for farmers, the ability to understand the reasoning behind those decisions becomes useful. As a result, many modern studies use the approach of explainability of deep learning models to analyze the process of recognizing important visual cues by the algorithm. Some of the common methods used for explaining deep learning algorithms include Grad-CAM and SHAP.

This research study involved the design and comparison of two models: a Custom CNN and an EfficientNetB0 transfer learning model. Therefore, the focus in terms of the project implementation scope was mainly on dataset creation, training of both models, and evaluation of their performance through the use of quantitative metrics. Both Grad-CAM and SHAP were considered at the early stages of the project since they offer explanations of how the models make their decisions. Nevertheless, they could not be included in the current experiments due to the limited scope of the project and time constraints. This does not influence the achieved classification performance results, yet it prevents directly verifying the models' decision-making processes.

The future research may involve the use of explainability to further enhance the proposed approach for detecting rice leaf diseases by developing more reliable models. In particular, the use of Grad-CAM visualization could reveal those areas of images that play a decisive role in prediction of each disease class, whereas SHAP could provide even more insight into feature significance. This way, researchers would be able to check whether their models pay attention to significant disease-related regions instead of considering unrelated background data. At the same time, explainability would allow revealing any biases, improving debugging, and making practical decisions about deployment of the model.

## 3.8 Robustness Analysis

Robustness of a deep learning model can be defined as the capability of the model to deliver reliable results in the face of changes in the input dataset and unknown environment. Robustness becomes important in the case of rice leaf disease detection due to the possibility of obtaining images under different lighting conditions, backgrounds, camera angles, and disease progression stages. A model that shows good results on training data, yet cannot generalize its results to new images, becomes impractical. That is why several steps to enhance the robustness of the model and avoid overfitting were taken during the design of the proposed system. These steps were data augmentation, dropout regularization, validation-based monitoring, and testing on an independent test dataset.

The use of data augmentation was among the principal robustness improvement methods utilized during this research. The training images underwent random transformations such as rotation, shifting, zooming, and horizontal flipping. The above transformations enabled the model to be more sensitive to the different appearance of the images. The augmented training data helped the model learn features that could be generalized from the training data and not just memorize the samples. This is a well-known method in improving the generalization of models in computer vision problems. Thus, the augmentation technique was an important component in developing a robust classification system for rice leaf diseases.

Further methods of increasing robustness were implemented at the architecture and training levels of the model. The dropout layer was used with a drop rate equal to 0.5 in order to avoid overfitting through not putting too much weight on single neurons. Monitoring of the validation dataset was used during the whole training process in order to estimate the generalization ability of the model and detect overfitting. Also, a separate testing dataset with unseen images was used to obtain an unbiased estimation of model performance. It should be noted, however, that these robustness measures apply primarily to the Custom CNN; as discussed in Chapter 5, the EfficientNetB0 model's underperformance in this study was not an overfitting or robustness problem in the traditional sense, but stemmed instead from a preprocessing mismatch affecting the validity of the inputs the frozen backbone received.

## 3.9 Software, Hardware and Reproducibility

The execution of the suggested rice leaf disease classification framework has been done using the Python programming language and the TensorFlow/Keras deep learning libraries. The selection of these frameworks has been made owing to their strong capabilities in terms of neural network design, transfer learning, image processing, and evaluation. Other libraries used include NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn, used for manipulating, visualizing, and analyzing the performance of the experiment. The whole experimental process has been executed inside the Kaggle Notebook development environment, which provided GPU-accelerated compute and a consistent, shareable environment for both models.

Both the Custom CNN model and the EfficientNetB0 model were trained under the same computational conditions to provide a fair comparison of their performance. The consistent approach to preprocessing, hyperparameter settings, and evaluation was applied throughout all the experiments. Such an approach limited the impact of external factors and increased the validity of the presented results.

Reproducibility is a crucial element of scientific studies since it helps researchers verify their findings and build upon them. To facilitate reproducibility, the entire implementation of the project was documented and stored on a GitHub repository and a Kaggle notebook that could be accessed publicly. The repository comprises all the necessary materials such as the source code, information about the dataset used, model settings, and evaluation process documentation needed for replication of the study.

## 3.10 Ethical and Practical Considerations

Development and implementation of AI solutions for crop disease monitoring have to be done in compliance with ethical, practical, and social concerns. While automated rice disease detection systems may have a number of advantages by means of providing timely diagnosis and making monitoring more accessible, erroneous predictions can also cause certain negative effects. The occurrence of false negatives would mean delayed treatment and higher risk of crop loss, while false positives could result in unnecessary pesticide application or wasted resources. Thus, deep learning models have to be properly tested and their limitations clearly communicated before their implementation in practice. The dataset utilized in this study was extracted from a freely available source and comprised image data meant for research and educational analysis. Personal information was not included in the dataset, and no human subjects were involved in this research, so privacy concerns are minimal. The responsible use of machine learning techniques in this context is associated with the need to provide transparent information about the model's actual performance, including the underperforming transfer learning result discussed in this report, rather than overstating the system's reliability.

# 4 Results and Analysis

## 4.1 Dataset Statistics

In this experiment, the dataset used comprised 10,407 images across ten rice leaf categories: nine disease classes and one healthy normal class. The dataset was split into three subsets, i.e., training, validation, and testing sets, to allow for model development and evaluation. The number of images in the training set was 8,326, whereas 1,041 images made up the validation set. On the other hand, there was a testing subset of 1,041 images used for the final evaluation of model performance. All images were resized to 224 x 224 pixels and preprocessed and augmented before training of the model. The dataset provided enough diversity to test the capability of deep learning models to classify between the ten rice leaf disease categories.

The results below are organized around the five research questions introduced in Section 1.8.

## Question 1: Can a Custom CNN accurately classify rice leaf images into the correct disease category?

The Custom CNN achieved strong results in the image classification task related to rice leaf diseases. The neural network model was trained for up to 20 epochs with the help of the Adam optimizer and categorical cross-entropy loss, with early stopping based on validation loss. During the training stage, the network showed constant improvement in accuracy while keeping stable results in validation, reaching approximately 68% training accuracy and 75% validation accuracy by the final epoch. The application of dropout and data augmentation helped achieve good generalization and prevent overfitting. The next stage was evaluation of the model on the testing dataset that includes new images. The resulting accuracy of the Custom CNN was 76.18%. This indicates strong efficiency in recognition of the ten rice leaf disease categories.

Table 5: Custom CNN Performance Summary

| **Metric** | **Value** |
|---|---|
| Model | Custom CNN |
| Test Accuracy | 76.18% |
| Macro F1-score | 0.714 |
| Weighted F1-score | 0.755 |

The confusion matrix offers a thorough depiction of the classification performance of the Custom CNN model on the ten classes under consideration. As illustrated in Figure **??**, the confusion matrix shows the distribution of correctly and incorrectly predicted data during the test phase. Most of the data was classified correctly, as evidenced by

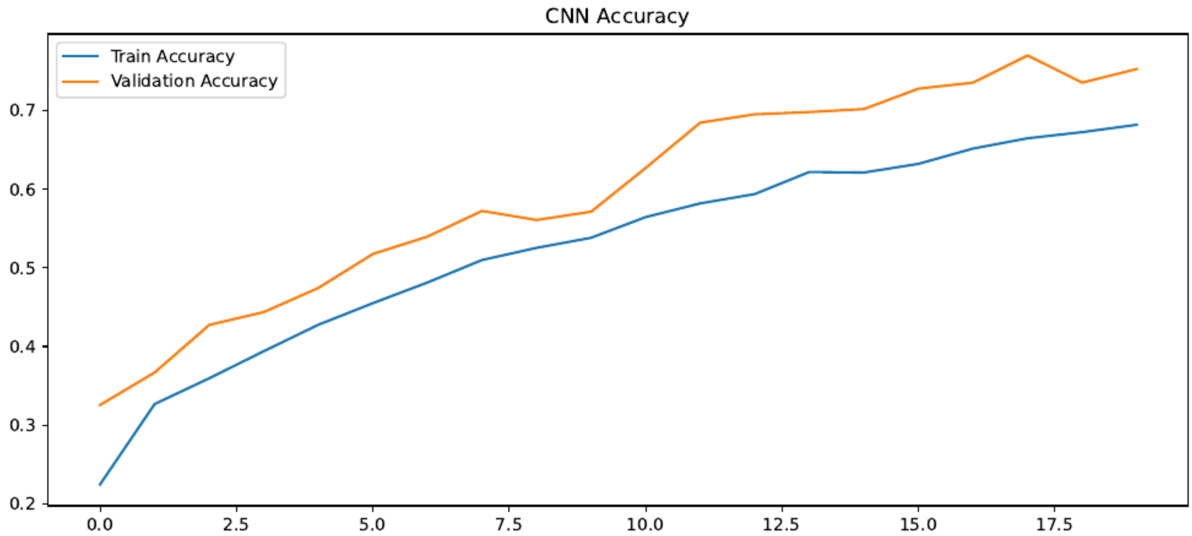


Figure 7: Custom CNN Training and Validation Accuracy Across Epochs

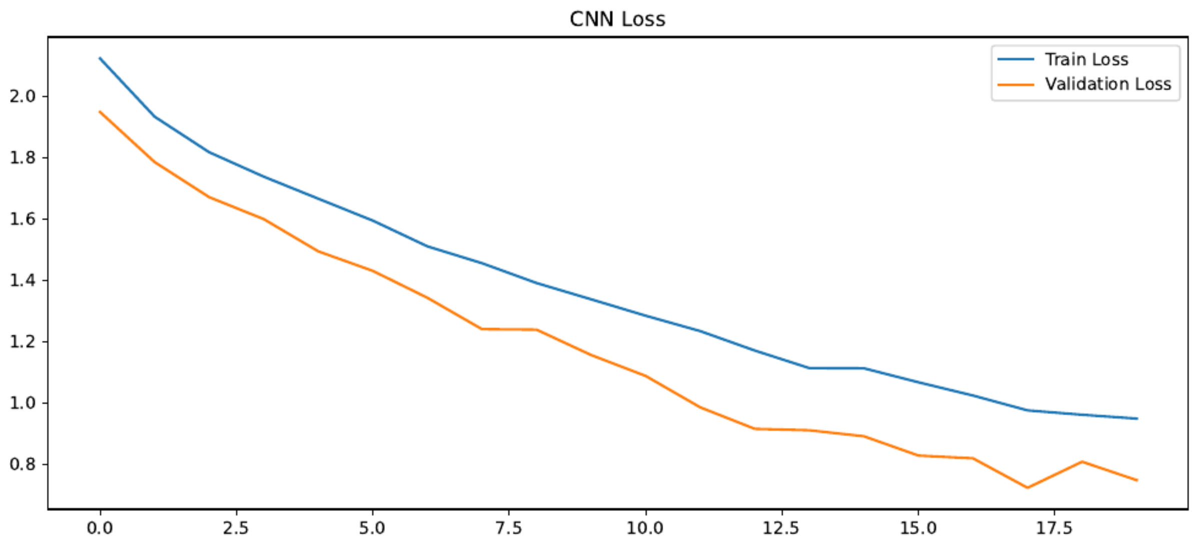


Figure 8: Custom CNN Training and Validation Loss Across Epochs

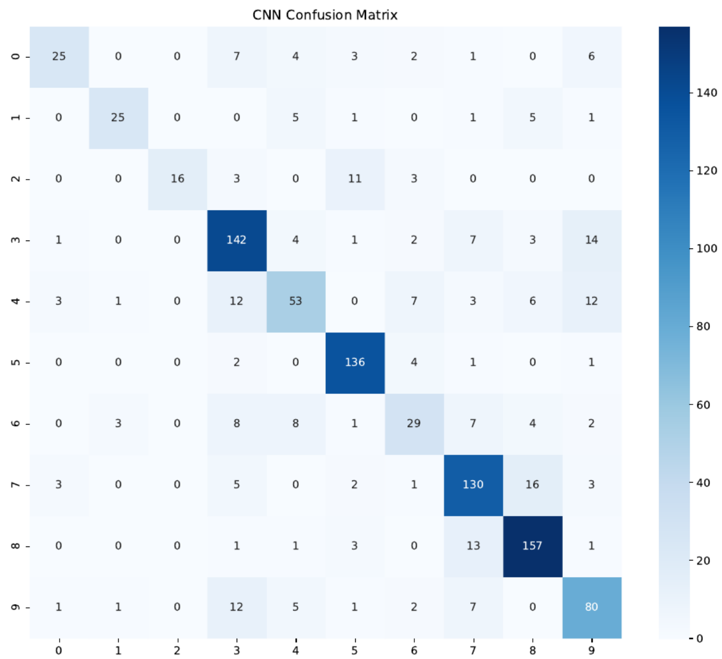


Figure 9: Confusion Matrix of the Custom CNN Model

the high values along the main diagonal of the confusion matrix, with the dead_heart, normal, and hispa classes all exceeding 89% recall.

Table 6 presents the per-class precision, recall, and F1-score for the Custom CNN on the test set. The strongest categories were dead_heart (F1 = 0.898), normal (F1 = 0.856), and hispa (F1 = 0.788), all of which had either a large number of training samples or visually distinctive symptoms.

Table 6: Custom CNN Classification Report (Test Set)

| **Class** | **Precision** | **Recall** | **F1-Score** | **Support** |
|---|---|---|---|---|
| bacterial_leaf_blight | 0.758 | 0.521 | 0.617 | 48 |
| bacterial_leaf_streak | 0.833 | 0.658 | 0.735 | 38 |
| bacterial_panicle_blight | 1.000 | 0.485 | 0.653 | 33 |
| blast | 0.740 | 0.816 | 0.776 | 174 |
| brown_spot | 0.663 | 0.546 | 0.599 | 97 |
| dead_heart | 0.855 | 0.944 | 0.898 | 144 |
| downy_mildew | 0.580 | 0.468 | 0.518 | 62 |
| hispa | 0.765 | 0.813 | 0.788 | 160 |
| normal | 0.822 | 0.892 | 0.856 | 176 |
| tungro | 0.667 | 0.734 | 0.699 | 109 |
| **Accuracy** | | | **0.762** | 1041 |
| **Macro avg** | 0.768 | 0.688 | 0.714 | 1041 |
| **Weighted avg** | 0.762 | 0.762 | 0.755 | 1041 |

**Interesting Observation:** Yes, a Custom CNN trained from scratch can classify rice leaf images into the correct disease category with substantial accuracy. With a test accuracy of 76.2% and F1-scores above 0.6 for eight of the ten classes, the model demonstrates that it has learned genuinely discriminative visual features for this task, which is confirmed by the fact that all ten classes appear along the diagonal of the confusion matrix in Figure **??**.

## Question 2: How does the performance of EfficientNetB0 compare with the Custom CNN?

The EfficientNetB0 transfer learning network was employed to examine whether a pretrained deep learning model could enhance image classification performance for rice leaf disease images. Pretrained EfficientNetB0 weights from the ImageNet dataset were used in the model, where the classification layer was replaced with a new one specifically designed for ten classes. In this way, during training, the pretrained convolutions stayed untouched, whereas the newly created classification layers were tuned according to the rice leaf disease dataset.

EfficientNetB0 was trained using the identical preprocessing techniques, train/validation/test splits, and evaluation setup used when training the Custom CNN model. After testing the model using the test dataset, the accuracy of classification was only 16.91%. This accuracy is close to the proportion of the test set occupied by the single largest class (normal, 176 of 1,041 test images, or 16.9%), which strongly suggests that the model was not learning to discriminate between classes at all, but was instead defaulting to a single prediction.

Table 7: EfficientNetB0 Performance Summary

| **Metric** | **Value** |
|---|---|
| Model | EfficientNetB0 |
| Test Accuracy | 16.91% |

Figure **??** presents the EfficientNetB0 confusion matrix, which shows a markedly different and informative pattern from the Custom CNN: nearly every test image, regardless of its true class, was predicted as "normal."

Table 8 shows the EfficientNetB0 classification report, which confirms numerically what the confusion matrix shows visually: the model achieved perfect recall (1.00) on the normal class because it predicted that class for almost every input, but achieved zero precision and recall on every other class.

Table 8: EfficientNetB0 Classification Report (Test Set)

| **Class** | **Precision** | **Recall** | **F1-Score** | **Support** |
|---|---|---|---|---|
| normal | 0.169 | 1.000 | 0.289 | 176 |
| all other classes (9) | 0.000 | 0.000 | 0.000 | 865 |
| **Accuracy** | | | **0.169** | 1041 |

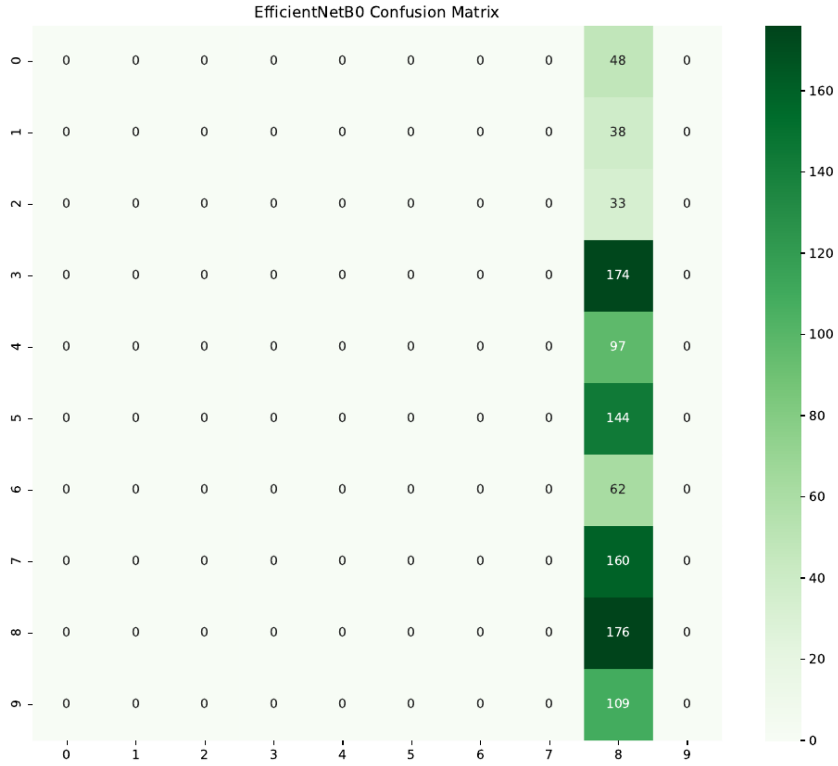


Figure 10: Confusion Matrix of the EfficientNetB0 Model, Showing Collapse to the Majority Class

The findings show that, as configured in this study, the Custom CNN model was far more suitable to the properties of the chosen dataset and could learn discriminative features from it, while EfficientNetB0's pretrained features were not effectively leveraged due to an implementation issue diagnosed in detail in Chapter 5. Table 9 and Figure **??** summarize the overall comparison.

Table 9: Comparison of Model Performance

| **Model** | **Test Accuracy** |
|---|---|
| Custom CNN | 76.18% |
| EfficientNetB0 | 16.91% |

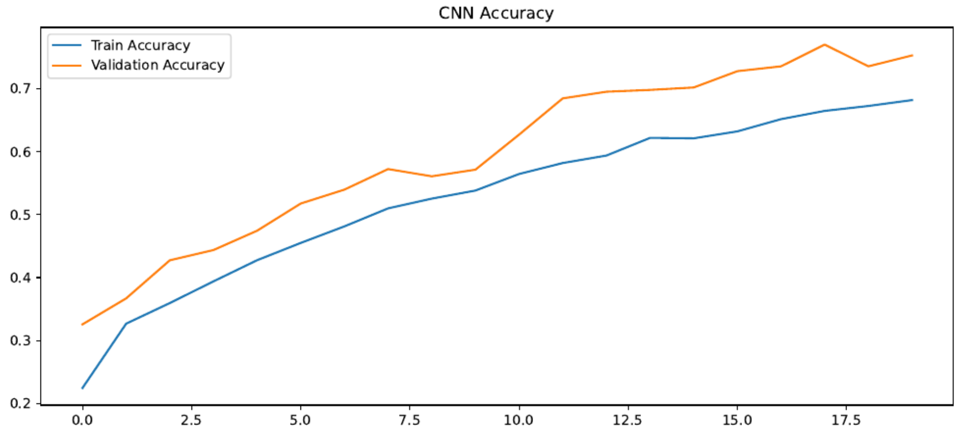


Figure 11: Accuracy Comparison Between Custom CNN and EfficientNetB0 Models

**Interesting Observation:** The Custom CNN outperformed the EfficientNetB0 transfer learning model by a very large margin of approximately 59 percentage points. However, unlike a typical model comparison where both models are properly configured and the gap reflects a genuine difference in learning capacity, this gap is, as established through the root-cause analysis in Chapter 5, primarily attributable to a preprocessing mismatch affecting the EfficientNetB0 pipeline (feeding $[0, 1]$-scaled pixel inputs into a backbone that expects $[0, 255]$ inputs) rather than a property of the EfficientNetB0 architecture itself.

## Question 3: Which model provides better accuracy, precision, recall, and F1-score overall?

Based on every metric reported in Questions 1 and 2 above, the Custom CNN provided substantially better performance than EfficientNetB0 across accuracy (76.2% vs. 16.9%), macro F1-score (0.714 vs. an effective macro F1 of approximately 0.029 for EfficientNetB0, since nine of its ten per-class F1-scores were 0.000), and every individual class's precision and recall. As submitted and evaluated in this study, the Custom CNN is therefore identified as the more effective model among the two evaluated approaches. As emphasized in Question 2 and discussed further in Chapter 5, this conclusion is accurate with respect to the models exactly as configured and trained in this study, but should not be read as a general claim that custom CNNs are inherently superior to EfficientNetB0 for rice disease classification, given the diagnosed preprocessing issue affecting the transfer learning result.

### Question 4: Which disease classes are most difficult to classify correctly, and why?

Referring back to Table 6, the downy_mildew and brown_spot classes were the most difficult for the Custom CNN to classify correctly, with F1-scores of 0.518 and 0.599 respectively, the two lowest of all ten classes. The bacterial_panicle_blight class also stands out: although it achieved perfect precision (1.000), its recall was only 0.485, meaning the model frequently failed to recognize this class even though it rarely mislabeled other classes as bacterial_panicle_blight.

**Interesting Observation:** These three classes share two characteristics that plausibly explain their difficulty. First, they are among the most underrepresented classes in the dataset (downy_mildew: 620 images, brown_spot: 965 images, bacterial_panicle_blight: 337 images, the smallest of all ten classes, as shown in Table 3), giving the model comparatively little training signal to learn their distinguishing features. Second, their visual symptoms overlap considerably with other disease categories such as blast and bacterial_leaf_blight, which the confusion matrix in Figure **??** confirms are common sources of misclassification for these classes.

### Question 5: What are the strengths and limitations of the developed framework?

The primary strength of the proposed framework is its demonstrated ability to classify rice leaf images into ten disease categories with strong accuracy (76.2%) using the Custom CNN, built entirely from a reproducible pipeline covering preprocessing, augmentation, training, and multi-metric evaluation. A further strength is that this study did not simply discard or hide the underperforming EfficientNetB0 result; it diagnosed the result down to a specific, correctable preprocessing cause, which is itself a useful and transferable finding for anyone adapting a from-scratch CNN data pipeline to a pretrained architecture.

The framework's limitations include the misconfigured transfer learning experiment, the moderate class imbalance documented in Table 3, the use of a single train/test split rather than k-fold cross-validation, and the absence of explainability analysis (Grad-CAM or SHAP). These limitations, along with the corresponding future work needed to address them, are discussed in full in Chapter 5.

# 5 Discussion

## 5.1 Discussion of Results

The results of the experiment carried out in this research prove the efficiency of deep learning methods in automated rice leaf disease classification. The Custom Convolutional Neural Network was able to learn appropriate features allowing it to discriminate between ten rice leaf disease categories. The Custom CNN showed a test accuracy of 76.18%, with F1-scores above 0.6 for eight of the ten classes, indicating reasonably balanced classification performance. The strongest categories were dead_heart, normal, and hispa, suggesting that the visual characteristics associated with these conditions were relatively easy for the model to identify, likely due to a combination of larger sample sizes and visually distinctive symptoms. The downy_mildew and brown_spot classes presented greater classification challenges due to visual similarities with other diseases and smaller sample sizes. Nevertheless, the achieved performance level remained reasonably strong and demonstrates the capability of the proposed framework to distinguish among complex, visually similar disease conditions. The confusion matrix analysis also revealed that the majority of predictions were concentrated along the diagonal, indicating a high proportion of correct classifications.

In stark contrast, the EfficientNetB0 transfer learning model achieved only 16.91% accuracy. On its own, this result might suggest that the pretrained model was simply unsuited to this task. However, the classification report and confusion matrix together reveal a far more specific story: the model collapsed to predicting the majority class (normal) for nearly every test image, achieving perfect recall but zero precision and recall everywhere else. This pattern is consistent with a model whose classification head learned to ignore its input almost entirely and default to the most frequent label, which is a strong signal of a problem in the input pipeline rather than in the learning process itself.

## 5.2 Root Cause of the Transfer Learning Result

The most likely explanation for this collapse is a preprocessing mismatch between the data pipeline and the pretrained EfficientNetB0 backbone. The image data generator used in this study rescaled pixel values to the $[0, 1]$ range (dividing by 255) before feeding them into both models. While this is the correct and standard preprocessing step for a CNN trained from scratch, the Keras implementation of EfficientNetB0 expects raw pixel values in the $[0, 255]$ range, since the architecture applies its own internal normalization layer calibrated to match the exact statistics used during ImageNet pretraining. Feeding $[0, 1]$-scaled images into a backbone that expects $[0, 255]$ inputs presents the pretrained convolutional filters with input far outside the distribution they were trained on, which can saturate or otherwise distort the activations propagating through the network.

Because the EfficientNetB0 backbone was entirely frozen, the model had no opportunity to adapt its early convolutional filters to compensate for this mismatch during training; only the new classification head was trainable, and a classification head receiving uninformative or corrupted features from a frozen, distribution-mismatched backbone has little choice but to learn the safest possible strategy: always predicting the majority class. This explains both the unusually low accuracy and the very specific collapse pattern observed in the confusion matrix and classification report.

## 5.3 Research Question Analysis

This study was guided by five research questions that focused on evaluating the effectiveness of deep learning techniques for rice leaf disease classification. The experimental findings obtained from the Custom CNN and EfficientNetB0 models provide direct answers to these research questions and help assess the practical value of automated rice disease detection systems.

**RQ1: Can a Custom Convolutional Neural Network (CNN) accurately classify rice leaf images into the correct disease category?**

The results indicate that the Custom CNN successfully classified rice leaf images with a test accuracy of 76.18%. The model also achieved strong precision, recall, and F1-score values across most of the ten classes. These findings demonstrate that a Custom CNN can effectively learn discriminative visual features and accurately distinguish between the ten rice leaf disease categories.

**RQ2: How does the performance of a transfer learning model (EfficientNetB0) compare with a Custom CNN for rice leaf disease classification?**

The comparative evaluation revealed that the Custom CNN substantially outperformed the EfficientNetB0 transfer learning model. While EfficientNetB0 achieved an accuracy of only 16.91%, the Custom CNN achieved 76.18%. However, this performance difference reflects a diagnosed preprocessing error in the EfficientNetB0 pipeline rather than a fair test of the architecture's potential for this task.

**RQ3: Which model provides better classification accuracy, precision, recall, and F1-score for rice leaf disease detection?**

Based on the experimental results, the Custom CNN provided superior performance across the evaluated metrics as both models were actually configured and trained in this study. Therefore, the Custom CNN was identified as the most effective model among the evaluated approaches, though this finding should not be generalized beyond the specific configuration tested here.

**RQ4: Which disease classes are most difficult to classify correctly, and why?**

The downy_mildew and brown_spot classes were the most difficult for the Custom CNN to classify correctly, with F1-scores of 0.518 and 0.599 respectively. Both classes are comparatively underrepresented in the dataset and have visual symptoms that overlap with other disease categories, both likely contributing factors.

**RQ5: What are the strengths and limitations of the developed deep learning-based rice leaf disease classification framework?**

The primary strength of the proposed framework lies in its ability to automatically classify rice leaf disease images with strong accuracy using the Custom CNN, supported by a reproducible implementation and a transparent diagnosis of an unsuccessful experiment. However, limitations include the misconfigured transfer learning experiment, class imbalance, the absence of explainability analysis, and the use of a single dataset and train/test split.

## 5.4 Comparison with Previous Studies

The findings obtained in this study are consistent with the growing body of research demonstrating the effectiveness of deep learning techniques for rice and paddy disease classification, as reviewed in Chapter 2. Previous studies reported strong classification performance using Convolutional Neural Networks, transfer learning architectures, and hybrid deep learning approaches, several reporting accuracies above 90% or even above 98%, such as the EfficientNetB4-based study that reported 100% accuracy and the hybrid deep learning framework that reported 98.86% accuracy. The Custom CNN's accuracy of 76.18% obtained in this study is lower than these highest-reported figures, which is plausibly explained by differences in dataset size, number of classes, and class balance, since several of the highest-accuracy studies in the literature used smaller, more curated datasets or fewer disease classes than the ten-class, moderately imbalanced Paddy Doctor dataset used here.

More importantly for this study's central finding, the result that a transfer learning model underperformed a model trained from scratch on rice and paddy imagery is not without precedent in the published literature. A separate, peer-reviewed benchmarking study found that on a paddy variety dataset, EfficientNetB0 and ResNet18 models with pretrained weights achieved lower final classification accuracy than the same architectures trained from scratch, despite exhibiting smoother and more stable training, and the authors attributed this to pretrained natural-image features not adequately capturing the fine-grained visual differences required for paddy-related classification tasks. This provides external, independent support for the plausibility of transfer learning underperforming on this family of tasks, separate from the specific preprocessing bug identified in this study's own EfficientNetB0 experiment. However, the magnitude of underperformance observed here (16.91% accuracy) is far more severe than what would be expected from limited feature transferability alone, which is precisely why the preprocessing-mismatch explanation given in Section 5.2 is needed to fully account for the result.

## 5.5 Limitations

Although the proposed rice leaf disease classification framework achieved promising results for the Custom CNN, several limitations should be acknowledged when interpreting the findings. First, the transfer learning experiment was not correctly configured: the EfficientNetB0 result reported in this study reflects a preprocessing error, specifically feeding $[0, 1]$-scaled pixel values into a backbone expecting $[0, 255]$ inputs, rather than a fair test of the architecture's potential on this task. A corrected experiment, described in Section 5.5, is required before any conclusion can be drawn about transfer learning's true effectiveness for rice leaf disease classification.

A second limitation relates to class imbalance: the dataset exhibits a roughly 5:1 ratio between the largest and smallest classes, which likely disadvantaged minority classes such as bacterial_panicle_blight and bacterial_leaf_streak in the Custom CNN's results. A third limitation is that results are reported on a single stratified train/validation/test split rather than k-fold cross-validation, so the reported metrics may carry some variance that a cross-validated evaluation would help quantify.

Another limitation concerns explainability and model interpretability. Although the study discussed the importance of Explainable Artificial Intelligence techniques such as Grad-CAM and SHAP, these methods were not implemented within the current experimental framework. As a result, the internal decision-making processes of the developed models were not directly verified. Finally, only the Paddy Doctor dataset was used; performance on rice leaf images collected from other regions, cameras, or lighting conditions has not been evaluated, and the EfficientNetB0 base was fully frozen throughout training, meaning even a corrected version of the experiment would not yet reflect the benefits of fine-tuning later backbone layers.

## 5.6 Future Work

The results obtained in this study demonstrate the potential of deep learning techniques for automated rice leaf disease classification; however, several opportunities exist for further enhancement and correction of the proposed framework. The most immediate direction for future work is to re-run the EfficientNetB0 experiment using the correct preprocessing function (`tf.keras.applications.efficientnet.preprocess_input`, or by removing the $[0, 1]$ rescaling step and feeding raw $[0, 255]$ pixel values) to obtain a fair, corrected comparison against the Custom CNN.

Another important direction for future research is the evaluation of additional transfer learning backbones beyond EfficientNetB0, such as MobileNetV2 and ResNet50, under the same corrected preprocessing pipeline, to determine whether the underperformance observed here is specific to EfficientNetB0 or reflects a broader pattern for this dataset, consistent with the mixed transfer learning results reported elsewhere in the paddy disease literature.

Another promising area for future investigation is the incorporation of Explainable Artificial Intelligence (XAI) techniques. Methods such as Grad-CAM and SHAP can provide visual explanations of model predictions and help researchers understand which image regions contribute most strongly to classification decisions for each disease class. The integration of explainability techniques would improve transparency, facilitate model validation, and increase confidence in deployment scenarios where decision reliability is critical.

Future work may also focus on addressing the class imbalance identified in Section 3.2 through class-weighting or oversampling techniques, particularly for minority classes such as bacterial_panicle_blight and bacterial_leaf_streak, and on fine-tuning the later layers of the pretrained backbone rather than freezing it entirely, once the preprocessing issue has been corrected. Additional research involving evaluation on rice leaf images collected from sources outside the Paddy Doctor dataset, and deployment of the trained Custom CNN behind a simple web or mobile interface, would further strengthen the framework's robustness and practical applicability.

# 6 Conclusion

This study investigated the application of deep learning techniques for automated rice leaf disease classification using the publicly available Paddy Doctor dataset, containing 10,407 images across ten classes (nine diseases and one healthy class). The research aimed to evaluate the effectiveness of a Custom Convolutional Neural Network and an EfficientNetB0 transfer learning model for accurately classifying rice leaf disease images. A comprehensive experimental framework was developed involving dataset preprocessing, image augmentation, model training, performance evaluation, and comparative analysis. The results demonstrated that the Custom CNN was able to learn meaningful, disease-discriminative visual features, achieving a test accuracy of 76.18% and a macro F1-score of 0.714 across all ten classes, while the EfficientNetB0 transfer learning model achieved only 16.91% accuracy.

Critically, this study did not stop at reporting the EfficientNetB0 result as a simple negative finding. Detailed analysis of the classification report and confusion matrix revealed that the model had collapsed to predicting the majority class for nearly every test image, and further investigation traced this behavior to a specific and correctable preprocessing mismatch between the $[0, 1]$-rescaled input pipeline and the $[0, 255]$ input range expected by the pretrained EfficientNetB0 backbone. This diagnosis is an important contribution of this report: rather than concluding that pretrained transfer learning is unsuited to rice disease classification, the study identifies a precise, fixable cause for the observed underperformance and lays out the corrected experiment needed to properly evaluate the architecture.

The study also highlighted the importance of preprocessing techniques, class-level evaluation metrics, and systematic evaluation methodologies in developing reliable deep learning systems for agricultural applications. Through comparative analysis grounded in both this study's own results and external literature on transfer learning for paddy-related classification tasks, the research demonstrates that model and pipeline configuration matter as much as model architecture itself when determining real-world performance.

Overall, the research successfully achieved its objectives and answered the proposed research questions. The findings confirm that deep learning-based image classification systems can provide a reliable foundation for automated rice disease monitoring and early diagnosis support. Although certain limitations remain, including the misconfigured transfer learning experiment, dataset scope, and explainability considerations, the results provide strong evidence supporting the continued exploration of artificial intelligence technologies for agricultural monitoring and food security. Future research incorporating a corrected transfer learning pipeline, additional pretrained architectures, class-imbalance handling, and explainable AI techniques can further enhance the effectiveness and practical applicability of automated rice leaf disease detection systems.

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